Vikas Dwivedi

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Research Profile

Scientific machine learning for forward and inverse partial differential equations, emphasizing fast physics-informed solvers, domain decomposition for sharp-gradient problems, interpretable and compressed neural representations, and physics-informed computer vision. Applications include computational fluid dynamics, biomedical imaging, and tropical cyclone intensity estimation.

Academic Appointments

Nov 2024 – present	**Postdoctoral Researcher	CREATIS, INSA Lyon, France** Biophysics-informed medical imaging and scientific machine learning.
Jul 2023 – May 2024	**Postdoctoral Researcher	University at Albany, SUNY, USA** Machine-learning approaches to tropical cyclone intensity estimation.
Jul 2022 – Mar 2023	**Postdoctoral Researcher	Robert Bosch Centre for Data Science and AI, IIT Madras, India** Physics-informed medical image segmentation.

Education

Jan 2015 – Mar 2021	**Ph.D., Mechanical Engineering	Indian Institute of Technology Madras** Thesis: *Physics Enhanced Machine Learning Methods for Flow Simulation*. Supervisor: Prof. Balaji Srinivasan. Awarded Best PhD Thesis in Data Science (2021).
Jul 2012 – Nov 2014	**M.Tech., Applied Mechanics	Indian Institute of Technology Delhi** Grade: 8.4.
Jul 2007 – May 2011	**B.E., Mechanical Engineering	IET, Devi Ahilya Vishwavidyalaya, Indore** Grade: 70%.

Industry Experience

Apr 2020 – Jun 2022	**Data Scientist	VuNet Systems, Bengaluru, India** Machine learning analysis of business and performance observability platforms.

Awards & Distinctions

2021	**Best PhD Thesis Award in Data Science**, Indian Institute of Technology Madras.
Master's / Ph.D.	MHRD scholarship during graduate studies.
2011	Graduate Aptitude Test in Engineering (Mechanical): All-India Rank **325** among 81,175 candidates.

Research Publications

Peer-reviewed journal and conference outputs are separated from workshop contributions. Author names and links reflect the verified publication list; see DOI or PDF links for source records.

Journal Articles (10)

[J1] V. Dwivedi, B. Srinivasan, M. Sigovan, and B. Sixou. “KAPI-LM: Kernel-Adaptive Physics-Informed Learning Machines for PDEs”. In: *Neural Networks* 206, 109740 (2026). DOI.

[J2] V. Dwivedi, M. Sigovan, and B. Sixou. “Meta-SPINN: Meta-Learned Basis Adaptation for Parametric PDEs”. In: *Machine Learning: Science and Technology* 7.4, 045060 (2026). DOI.

[J3] V. Dwivedi, E. Schiassi, B. Sixou, and M. Sigovan. “Gated X-TFC: Soft Domain Decomposition for Forward and Inverse Problems in Sharp-Gradient PDEs”. In: *Neurocomputing* 676, 133090 (2026). DOI.

[J4] V. Dwivedi, B. Sixou, and M. Sigovan. “Curriculum Learning-Driven PIELMs for Fluid Flow Simulations”. In: *Neurocomputing* 650, 130924 (2025). DOI.

[J5] V. Dwivedi, B. Srinivasan, and G. Krishnamurthi. “Physics Informed Contour Selection for Rapid Image Segmentation”. In: *Scientific Reports* 14.1, 6996 (2024). DOI.

[J6] V. Dwivedi and B. Srinivasan. “A Normal Equation-Based Extreme Learning Machine for Solving Linear Partial Differential Equations”. In: *Journal of Computing and Information Science in Engineering* 22.1, 014502 (2022). DOI.

[J7] V. Dwivedi, N. Parashar, and B. Srinivasan. “Distributed Learning Machines for Solving Forward and Inverse Problems in Partial Differential Equations”. In: *Neurocomputing* 420 (2021), pp. 299–316. DOI.

[J8] V. Dwivedi and B. Srinivasan. “Solution of Biharmonic Equation in Complicated Geometries with Physics Informed Extreme Learning Machine”. In: *Journal of Computing and Information Science in Engineering* 20.6, 061004 (2020). DOI.

[J9] V. Dwivedi and B. Srinivasan. “Physics Informed Extreme Learning Machine (PIELM) — A Rapid Method for the Numerical Solution of Partial Differential Equations”. In: *Neurocomputing* 391 (2020), pp. 96–118. DOI.

[J10] V. Dwivedi and B. Srinivasan. “Prediction of Transition in Temporal Mixing Layer Using ILES”. In: *International Journal of Advances in Engineering Sciences and Applied Mathematics* (2018). DOI.

Conference Proceedings (2)

[C1] A. G. Srinivasan, V. Dwivedi, and B. Srinivasan. “Towards Sustainable Scientific Machine Learning: Fast and Interpretable PDE Solvers via RBF-PIELM”. In: *Proceedings of the 13th ACM IKDD International Conference on Data Science (CODS ’25)*. 2025, pp. 376–386. DOI.

[C2] V. Dwivedi. “Global Versus Local: Evaluating AlexNet Architectures for Tropical Cyclone Intensity Estimation”. In: *International Conference on Pattern Recognition*. 2024, pp. 409–423. DOI.

Workshop Papers (5)

[W1] A. Daniel, V. Dwivedi, and B. Srinivasan. “AB-PIELMs: Adaptive-Basis Physics-Informed Extreme Learning Machines for Residual-Driven Domain Decomposition”. In: *ICLR 2026 Workshop on AI and Partial Differential Equations (AI & PDE)*. 2026. PDF.

[W2] V. Dwivedi, M. Sigovan, and B. Sixou. “Kernel-Adaptive Physics-Informed Shallow Meta-Learning for Parametric Linear PDEs”. In: *ICLR 2026 Workshop on AI and Partial Differential Equations (AI & PDE)*. 2026. PDF.

[W3] A. G. Srinivasan, V. Dwivedi, and B. Srinivasan. “Deep vs. Shallow: Benchmarking Physics-Informed Neural Architectures on the Biharmonic Equation”. In: *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences (ML4PS)*. 2025. PDF.

[W4] V. Dwivedi, M. Sigovan, and B. Sixou. “Curriculum-Learning PIELMs for Hemodynamic Flows”. In: *NeurIPS 2025 Workshop on Optimization for Machine Learning (OPT)*. 2025. PDF.

[W5] V. Dwivedi, M. Sigovan, and B. Sixou. “Empirical-Bayes XTFC for Inverse Parameter Estimation”. In: *NeurIPS 2025 Workshop on Optimization for Machine Learning (OPT)*. 2025. PDF.

Manuscript Under Review

2026 — **CFKS: Continuum-Feature Kernel PDE Solvers**. V. Dwivedi, B. Srinivasan, B. Sixou and M. Sigovan. Submitted to *Journal of Computational and Applied Mathematics*. *Under review; not a published journal article.*

Invited Talks & Teaching

Nov 2023	**Invited lecture**: “Scientific Machine Learning for Fluid Mechanics,” IIT Kanpur, India.
2022	**Keynote speaker**, International Conference in Applied Mechanics, Machine Learning and Applied Computation, NIT Raipur, India.
NPTEL	Taught a short-term course on **Data-Centric AI**.

Selected Additional Conference Presentations

Dec 2020	“Numerical experiments to solve partial differential equations with a denoising auto-encoder,” 8th International Conference on Fluid Mechanics and Fluid Power, IIT Guwahati, India.
Dec 2019	“Rapid solution of PDEs in complex geometries with physics informed extreme learning machine,” 1st Indo-German Conference on Computational Mathematics, IISc Bengaluru, India.
Sep 2019	“Data driven solution of linear diffusion equation with distributed physics informed neural networks,” 7th Asian Symposium on Computational Heat and Mass Transfer, Tokyo University of Science, Japan.

Academic Service

Journal reviewer: *Neurocomputing*; *Scientific Reports*; *Applied Mathematical Modelling*.

Referees

Prof. Balaji Srinivasan sbalaji@dsai.iitm.ac.in
Professor, Mechanical Engineering and Wadhwani School of Data Science and AI, IIT Madras, India

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Prof. Bruno Sixou bruno.sixou@insa-lyon.fr
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Co-Founder and CTO, VuNet Systems, Bengaluru, India

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